

SQL CHEAT SHEET — Option A (Clean & Simple)

TABLES:

genre(genre_id PK, genre_name)

album(album_id PK, album_title, artist, release_year, popularity, revenue, rating)

album_genres(album_id FK, genre_id FK)

BASIC SELECT:

SELECT * FROM album;

SELECT album_title FROM album ORDER BY album_title;

DISTINCT:

SELECT DISTINCT artist FROM album;

WHERE FILTERS:

SELECT * FROM album WHERE rating > 4;

SELECT * FROM album WHERE artist IN ('Drake','Taylor Swift');

ORDER BY:

SELECT * FROM album ORDER BY popularity DESC;

SELECT * FROM album ORDER BY release_year ASC;

LIMIT / OFFSET:

SELECT * FROM album ORDER BY rating DESC LIMIT 5;

SELECT * FROM album ORDER BY revenue DESC LIMIT 1 OFFSET 2;

BETWEEN:

SELECT * FROM album WHERE popularity BETWEEN 50 AND 80;

JOINS (Many-to-Many):

SELECT a.album_title, g.genre_name

FROM album a

JOIN album_genres ag ON a.album_id = ag.album_id

JOIN genre g ON ag.genre_id = g.genre_id;

LEFT JOIN:

SELECT a.album_title, g.genre_name

```
FROM album a
LEFT JOIN album_genres ag ON a.album_id = ag.album_id
LEFT JOIN genre g ON ag.genre_id = g.genre_id;
```

AGGREGATION:

```
SELECT artist, COUNT(*) AS num_albums
FROM album
GROUP BY artist;
```

AVG, MAX, MIN:

```
SELECT AVG(rating) FROM album;
SELECT MIN(popularity), MAX(popularity) FROM album;
```

HAVING:

```
SELECT artist, AVG(rating) AS avg_rating
FROM album
GROUP BY artist
HAVING avg_rating > 7;
```

CASE WHEN:

```
SELECT album_title, rating,
CASE
WHEN rating < 5 THEN 'low'
WHEN rating < 8 THEN 'medium'
ELSE 'high'
END AS rating_level
FROM album;
```

GENRE AGGREGATION:

```
SELECT g.genre_name, COUNT(a.album_id)
FROM genre g
JOIN album_genres ag ON g.genre_id = ag.genre_id
JOIN album a ON ag.album_id = a.album_id
GROUP BY g.genre_name;
```